
INVERSE FOLDING MODELS TRACK STRUCTURAL COMPATIBILITY IN pMHC-TCR COMPLEXES

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ABSTRACT

Predicting cross-reactivity in T-cell receptor (TCR)-based cancer therapies is a major challenge. Previous clinical studies failed to identify cross-reactive targets, ultimately resulting in fatal cardiac toxicity in two patients (Linette et al., 2013; Cameron et al., 2013), and the combinatorial vastness of the promiscuous peptide-MHC (pMHC) landscape is beyond the reach of experimental sampling. While inverse folding (IF) models can sample peptide sequences from a distribution conditioned on structure, whether they recover the native peptide or map the tolerated neighborhood underlying cross-reactivity is unknown. Here we sampled peptides from four IF models across 43 pMHC-TCR crystal structures and scored them for predicted presentation and structural compatibility. We show that IF models do not recover the native peptide or rank binding strength, but instead sample from the sequence space compatible with the structural context they are conditioned on, offering a handle on the tolerated neighborhood that drives cross-reactivity. Together, these results establish IF model sampling distributions as an effective probe of the constraints governing peptide recognition and cross-reactivity, critical for TCR-based therapies.

1 INTRODUCTION

T-cell receptor (TCR)-based therapies have shown promising results for cancer treatment (Hong et al., 2023; Chen and Bulek, 2024). However, cross-reactivity poses a difficult challenge, as previous clinical studies have failed to identify cross-reactive targets, ultimately resulting in fatal cardiac toxicity in two patients (Linette et al., 2013; Cameron et al., 2013). Cross-reactive peptide-MHC-TCR (pMHC-TCR) complexes remain a fundamental barrier to scaling these personalized medicine approaches. Experimental methods sample a narrow slice of the pMHC landscape, and the combinatorially vast nature of the promiscuous pMHC is beyond the reach of experimental sampling (Mason, 1998; Wooldridge et al., 2012). Computational approaches have shown promise in evaluating potential pMHC interactions, whereas TCR context poses a more difficult question. Previous work has shown that pMHC-TCR experimentally resolved structures provide a grounded structural prior for sequence recovery of the CDR3 interface (Ribeiro-Filho et al., 2024). Inverse folding (IF) models sample sequences from a distribution conditioned on the structure (Dauparas et al., 2022; Hsu et al., 2022), and across 38 pMHC-TCR complexes, it was shown to recover the CDR3 interface residues (Ribeiro-Filho et al., 2024), indicating that IF models capture non-trivial distributional shifts from the pMHC-TCR interface. In this work, we ask whether the IF models recover the native peptide of the crystal structure, whether IF models capture the sequence distribution that is compatible with the interface which would conversely be compatible with cross-reactive sequences, and whether there is a shift in the context of the TCR reshapes the sequence sampling distribution. We therefore ask if IF models recover the native peptide, and whether the sample sequence space concentrates on the tolerated neighborhood underlying cross-reactivity.

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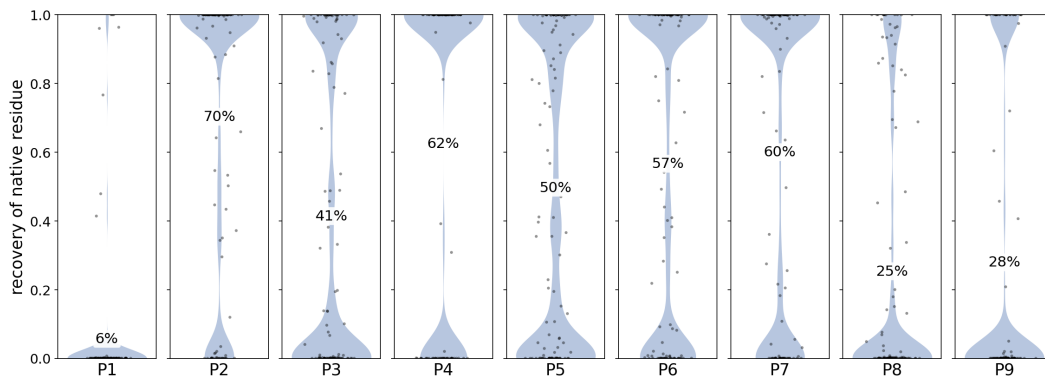


Figure 1: Per-position recovery of the native residue, 30 nonamer crystals, four models, pMHC+TCR context.

2 MATERIALS AND METHODS

2.1 PANEL AND MODELS

We downloaded 43 pMHC-TCR crystal structures from the Protein Data Bank (Berman et al., 2000) (Supplementary Table S1). Every structure includes a bound peptide, the MHC chains, and a complete TCR $\alpha\beta$ heterodimer (Bjorkman et al., 1987) (Supplementary Figure S1). Out of these, 40 structures are HLA-A*02:01 and the remaining three are one HLA-B*51:01 (4MJI), one HLA-B*08:01 (1MI5), one HLA-B*35 (2AK4). The average crystallographic resolution is 2.48 Å (Supplementary Figure S2). This set has 29 distinct peptides and 22 distinct CDR3 β sequences. Because SLLMWITQC is solved in five crystals across two closely related 1G4 variants, we also report a non-redundant subset of 30 HLA-A*02:01 structures over 26 peptides, keeping the best-resolution member of each TCR family and peptide group (Supplementary Figure S3). We apply four models: ProteinMPNN (Dauparas et al., 2022), ProteinMPNN (no MHC), the same model architecture retrained with MHC/TCR-class structures excluded, ESM-IF1 (Hsu et al., 2022) and LigandMPNN (Dauparas et al., 2025) (Supplementary Table S2). For every run, we only sample the peptide chain. Additionally, as an ablation experiment, we removed the TCR from the same structures and for each condition was sampled to 10,000 sequences with temperature of $T = 0.1$. For pMHC binding affinity prediction we apply MHCflurry (O’Donnell et al., 2020) and ESMCBA (EvolutionaryScale Team, 2024; Mares et al., 2025), and set a threshold of 500 nM for strong binders (Sette et al., 1994).

3 RESULTS

3.1 MODELS SAMPLE A STRONGLY POSITION-BIASED SEQUENCE SPACE

To establish how much of the peptide sequence space these models explore, we sampled 10,000 designs per structure and model combination in the pMHC-TCR context at $T=0.1$. All four models sample on average 49 unique sequences per structure, a highly restricted space (Supplementary Figure S4). Per-position Shannon entropy shows that every model imposes strong position-specific bias (Supplementary Figures S5–S9).

3.2 THE TWO MHC ANCHORS IMPOSE DIFFERENT CONSTRAINTS AND RECOVER DIFFERENTLY

As shown in prior work, IF models sample a sequence-distribution conditioned on the structure and have been shown to identify functionally critical residues (Dauparas et al., 2022; Hsu et al., 2022; Ribeiro-Filho et al., 2024; Deng et al., 2025; Luo et al., 2023). We ask whether critical positions (P2, P Ω) of the pMHC-TCR shape the recovery of the native sequence across the 30 nonamer structures and 4 models. We find that recovery is biased in an unexpected way: P2, seated in pocket B, constrains a specific side chain and recovers well above the interior average (70% vs 49% interior,

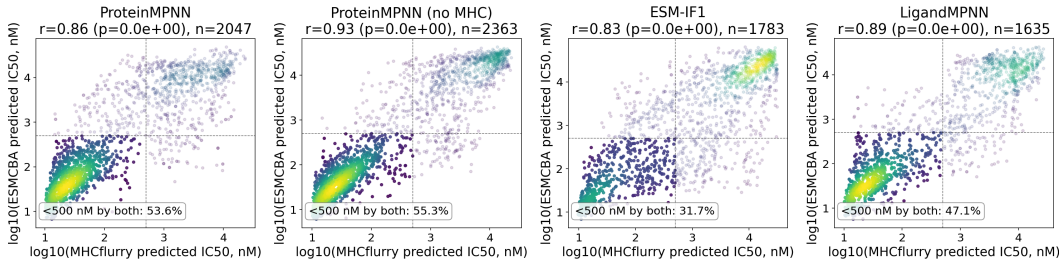


Figure 2: MHCflurry vs. ESMCBA predicted IC50 for every unique design, one panel per model.

$p = 5.2 \times 10^{-6}$), whereas P Ω , seated in pocket F, constrains the backbone but not side-chain identity and recovers significantly below it (28% vs 49%, $p = 7.5 \times 10^{-7}$) (Falk et al., 1991; Madden, 1995) (Figure 1). Both are structurally critical for pMHC binding, yet only P2 is constrained to a single side chain. On the non-redundant HLA-A*02:01 set (one structure per TCR family and per peptide, 30 structures over 26 peptides) P2 still recovers above interior ($p = 0.010$) and P Ω still falls below it ($p < 10^{-4}$) (Supplementary Figure S3). Canonically, HLA-A*02:01 anchor residues are aliphatic non-polar (Rammensee et al., 1999). The native peptides’ anchor residues are more hydrophobic than their own interior (Kyte-Doolittle scale (Kyte and Doolittle, 1982), $p < 0.0001$ for both), and the designs reproduce that chemistry without recovering the native residue ($p < 10^{-10}$ for both) (Supplementary Figure S10).

To evaluate if the recovery of the native sequence is a function of the structure’s resolution, we first pooled across the 43 structures, where the correlation does not reach significance ($r = -0.23$, $p = 0.14$), and only ESM-IF1 shows a relationship individually ($r = -0.37$, $p = 0.016$; Supplementary Figure S11). We then used five structures (2P5W, 2BNR, 2F53, 2F54 and 2PYE) that are independent crystals of the identical peptide (SLLMWITQC, the NY-ESO-1 epitope (Chen et al., 2005; Sami et al., 2007)) differing primarily in resolution (1.9–2.7 Å). Declining resolution decreases recovery for ProteinMPNN (no MHC) ($r = -0.97$, $p = 0.005$) and, less clearly, ESM-IF1 ($r = -0.84$, $p = 0.075$), and increases the number of unique designs for both ProteinMPNN variants ($r = 0.97$, $p = 0.006$ and $r = 0.95$, $p = 0.012$; Supplementary Figure S12). LigandMPNN shows the opposite trend ($r = +0.19$ and $r = -0.20$, both n.s.).

3.3 MOST PEPTIDES SAMPLED BY IF MODELS ARE PREDICTED TO BE PRESENTED

We asked whether those sampled peptides would be presented, scoring every unique design from the 40 HLA-A*02:01 structures for predicted HLA-A2 binding with two unrelated predictors, MHCflurry (O’Donnell et al., 2020) and ESMCBA (EvolutionaryScale Team, 2024; Mares et al., 2025), neither of which had seen this panel (Figure 2). The two predictors correlate strongly on the sampled designs ($r = 0.83$ – 0.93 per model in full context, 0.82 – 0.93 without the TCR) and cluster similarly in a UMAP projection of the ESMCBA embeddings (McInnes et al., 2018) (Supplementary Figure S13). We find that ProteinMPNN and LigandMPNN consistently sample higher predicted affinity than ESM-IF1. Below the 500 nM cutoff by both predictors (Sette et al., 1994), 47–55% of ProteinMPNN-family and LigandMPNN designs pass, compared with 32% for ESM-IF1 ($\chi^2 = 266.64$, $p = 2 \times 10^{-57}$).

3.4 WITHOUT THE TCR, IF MODELS SAMPLE MORE WIDELY BUT LESS CONFIDENTLY

Given that the models sample sequences compatible with the pMHC groove, we then evaluated the effect of the TCR on what these models propose. We performed an ablation experiment, resampling every structure without the TCR from the input. Every model explores substantially more diverse sequence space without the TCR, with unique design counts roughly doubling or tripling (ProteinMPNN 2121 $\rightarrow$ 6049; ProteinMPNN (no MHC) 2434 $\rightarrow$ 4070; ESM-IF1 1824 $\rightarrow$ 3539; LigandMPNN 1641 $\rightarrow$ 4343; Supplementary Figure S14). This larger space skews toward higher predicted affinity, but not uniformly across models. ProteinMPNN (no MHC) shows the largest shift (strong-binder rate by both predictors 55% $\rightarrow$ 81%), and LigandMPNN a moderate one (47% $\rightarrow$ 56%), while ProteinMPNN and ESM-IF1 change less (54% $\rightarrow$ 60% and 32% $\rightarrow$ 38%) (Supplementary Fig-

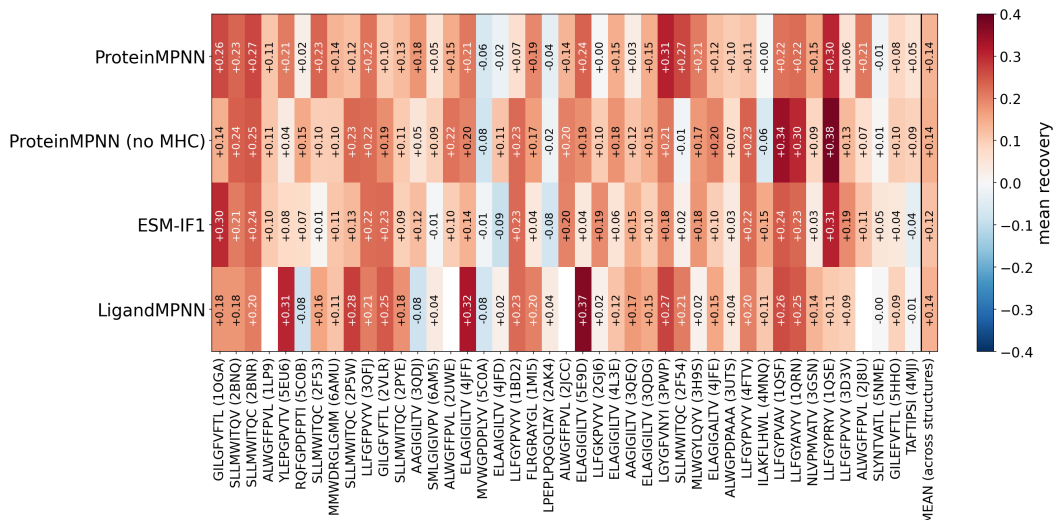


Figure 3: Recovery with the TCR minus recovery without, per crystal and model, ordered by resolution.

ure S15). Recovery moves the other way, falling from 44%-48% to 29%-36%, and the size of that loss varies by crystal (Figure 3).

Leveraging the models’ negative log likelihood (NLL) as a confidence metric for the sequences, we explore the distributional shifts in NLL induced by the TCR context. ProteinMPNN and ProteinMPNN (no MHC) report an average per-residue NLL for each design, where lower values indicate higher confidence; LigandMPNN reports an overall confidence score, for which higher values indicate higher confidence. Across the HLA-A*02:01 structures, removing the TCR shifts every model to a lower confidence (ProteinMPNN mean $0.845 \rightarrow 1.047$; ProteinMPNN (no MHC) $0.741 \rightarrow 0.904$; LigandMPNN $0.477 \rightarrow 0.401$; Mann-Whitney $p < 10^{-200}$ for all three; Supplementary Figure S16). Per-structure analysis shows the shift reaches significance in almost every structure (43 of 43 for ProteinMPNN, 41 of 43 for ProteinMPNN (no MHC), 38 of 39 for LigandMPNN; Supplementary Figure S17). Removing the TCR moves the sampled sequences into a nearly disjoint set that is predicted to be strong binders by both models. There is nearly no overlap (<2%) between the sequences sampled with and without the TCR, and the rate for <500 nM rises from 48% to 60%. The TCR interface shapes the sampling distribution toward the native epitope and far from the promiscuous pMHC sequence space.

3.5 RECOVERY TRACKS STRUCTURAL COMPATIBILITY, NOT BINDING AFFINITY

From the sequence recovery analysis, we find a bias at the P2 anchor, and the sampled sequences score as strong pMHC binders in TCR context. We next asked whether sequence recovery also serves as a proxy for functional pMHC-TCR interactions. We cross-referenced SKEMPI 2.0 (Jankauskaitė et al., 2019) and IEDB (Vita et al., 2019) against each structure’s crystallized TCR clone, matched by CDR3 sequence, and identified 21 (structure, position) pairs with an experimentally measured mutation that alters TCR binding. At these positions, designs recover the native residue at 54% with the TCR and 34% without, whereas the rest is at 44–48% and 29–36%. Then, we evaluate the question with the experimental binding affinities. Restricting to single mutants yields 23 complex/position pairs in SKEMPI across 11 human MHC class I complexes with measured $\Delta\Delta G$. At critical mutant positions (>1 kcal/mol), the recovery gain from TCR context is (+0.143 vs +0.111) not significant ($p = 0.26$, Mann-Whitney; Supplementary Figure S18). We next used the crystallized DMF5 receptor (PDB: 6AM5 and 6AMU), which presents the same HLA-A*02:01 with two different peptides and has been pivotal in defining cross-reactive ligands in pMHC-TCR (Riley et al., 2018). The same study characterized two 10-mers and 13 peptide variants measured for KD and thermal stability (Riley et al., 2018). The first, SMLGIGIVPV, which carries a central hydrophobic GIG core motif, and the second, MMWDRGLGMM, a DRG motif with a charged residue. We asked how the IF models

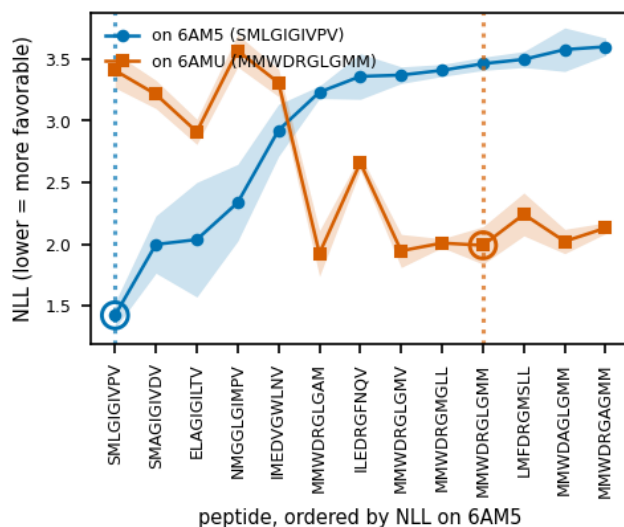


Figure 4: Thirteen DMF5 peptides scored on both backbones; rings mark each structure’s own peptide.

score and rank each peptide on its own backbone and on the others. Across all three scoring models, each structure’s own crystallized peptide ranks lower on average of 13 on its own backbone and 11 on average of 13 on the other (Figure 4), and the score therefore reads the backbone it is conditioned on. Across the nine peptides with a measured KD rather than a bound (four are reported as $>500 \mu\text{M}$ or not detected), the score carries no information about affinity on either backbone, for any of the three models ($|r| \leq 0.32$, all $p > 0.4$; Supplementary Figure S19). IF models preferentially score peptides that are compatible with a given groove and backbone, which is a quantity relevant to cross-reactivity, orthogonal to rank or binding strength.

The class-defining core is recovered well above the rest of the peptide. Across the six 10-mers carrying GIG at positions 4-6, the core recovers at 0.86 against an interior baseline of 0.44, positive in every structure ($p = 0.031$, paired Wilcoxon). The models therefore reproduce the characterized conserved core for this motif.

4 DISCUSSION

IF models have rapidly advanced structure-based protein design (Dauparas et al., 2022; 2025; Hsu et al., 2022), and their reported ability to recover native interface residues (Ribeiro-Filho et al., 2024) raises the question of whether they capture the biophysics of an interface or reproduce only its most constrained positions. Here we show that structural conditioning is not sufficient to recover the native peptide sequence, and that removing the TCR shifts both recovery and the distribution of sequences each model is confident in. The MHC anchors reveal mechanistic differences where pocket B constrains a single side chain, while pocket F constrains chemistry rather than side-chain identity. The residue recovery is observed in pocket B, and fails in pocket F. IF models sample a sequence distribution with strong predicted pMHC interactions, but the confidence scores reveal compatibility with the groove and backbone the model was conditioned on rather than the strength of binding. This work positions IF models as tools to characterize the sequence space compatible with a groove and a TCR. That compatible sequence space is precisely the tolerated neighborhood that gives rise to cross-reactivity. Their sampling distributions therefore serve as a probe of the constraints governing peptide recognition and cross-reactivity, which are critical for TCR-based therapies.

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A SUPPLEMENTARY FIGURES

PDB	Peptide	Length	Allele	Resolution (Å)
2P5W	SLLMWITQC	9	HLA-A*02:01	2.20
1QSF	LLFGYPVAV	9	HLA-A*02:01	2.80
1QRN	LLFGYAVYV	9	HLA-A*02:01	2.80
2BNR	SLLMWITQC	9	HLA-A*02:01	1.90
2GJ6	LLFGKPVYV	9	HLA-A*02:01	2.56
2F53	SLLMWITQC	9	HLA-A*02:01	2.10
2F54	SLLMWITQC	9	HLA-A*02:01	2.70
3QDG	ELAGIGILTV	10	HLA-A*02:01	2.69
3QEQ	AAGIGILTV	9	HLA-A*02:01	2.59
3QFJ	LLFGFPVYV	9	HLA-A*02:01	2.29
3GSN	NLVPMVATV	9	HLA-A*02:01	2.80
1OGA	GILGFVFTL	9	HLA-A*02:01	1.40
3UTS	ALWGPDPA AAA	10	HLA-A*02:01	2.71
5C0A	MVWGPDPLYV	10	HLA-A*02:01	2.46
5C0B	RQFGPDFPTI	10	HLA-A*02:01	2.03
5HHO	GILEFVFTL	9	HLA-A*02:01	2.95
5EU6	YLEPGPVTV	9	HLA-A*02:01	2.02
2VLR	GILGFVFTL	9	HLA-A*02:01	2.30
4MJI	TAFTIPSI	8	HLA-B*51:01	2.99
5NME	SLYNTVATL	9	HLA-A*02:01	2.94
1BD2	LLFGYPVYV	9	HLA-A*02:01	2.50
1LP9	ALWGFFPVL	9	HLA-A*02:01	2.00
1MI5	FLRGRAYGL	9	HLA-B*08:01	2.50
1QSE	LLFGYPRYV	9	HLA-A*02:01	2.80
2AK4	LPEPLPQGQLTAY	13	HLA-B*35	2.50
2BNQ	SLLMWITQV	9	HLA-A*02:01	1.70
2J8U	ALWGFFPVL	9	HLA-A*02:01	2.88
2JCC	ALWGFFPVL	9	HLA-A*02:01	2.50
2PYE	SLLMWITQC	9	HLA-A*02:01	2.30
2UWE	ALWGFFPVL	9	HLA-A*02:01	2.40
3D3V	LLFGFPVYV	9	HLA-A*02:01	2.80
3H9S	MLWGYLQYV	9	HLA-A*02:01	2.70
3PWP	LGYGFVNVI	9	HLA-A*02:01	2.69
3QDJ	AAGIGILTV	9	HLA-A*02:01	2.30
4FTV	LLFGYPVYV	9	HLA-A*02:01	2.74
4JFD	ELAAIGILTV	10	HLA-A*02:01	2.46
4JFE	ELAGIGALT	10	HLA-A*02:01	2.70
4JFF	ELAGIGILTV	10	HLA-A*02:01	2.43
4L3E	ELAGIGILTV	10	HLA-A*02:01	2.56
4MNQ	ILAKFLHWL	9	HLA-A*02:01	2.74
5E9D	ELAGIGILTV	10	HLA-A*02:01	2.51
6AM5	SMLGIGIVPV	10	HLA-A*02:01	2.39
6AMU	MMWDRGLGMM	10	HLA-A*02:01	2.15

Table S1: **The forty-three-structure panel.** PDB accession, native peptide, peptide length, MHC allele, and crystallographic resolution for every structure in the panel. The rule separates the twenty class I HLA-A*02:01-dominated structures above from the twenty-three added subsequently, all human MHC class I.

Model	Params	Architecture	Training data
ProteinMPNN	1.66M	Message-passing GNN	PDB (pdb_2021aug02), general backbones
ProteinMPNN (no MHC)	1.66M	Message-passing GNN (identical to above)	Same corpus, MHC/TCR-class structures excluded
ESM-IF1	141.7M	GVP-GNN + transformer	CATH-derived backbones
LigandMPNN	2.62M	Message-passing GNN + ligand context	PDB, augmented with explicit ligand/heteroatom context

Table S2: **Models compared.** Parameter counts verified directly from each checkpoint’s `state_dict`. ESM-IF1 is the only tool whose training set is built substantially from AlphaFold2-predicted structures (Jumper et al., 2021) rather than solved crystal structures (~12M predicted backbones (Hsu et al., 2022), on top of its CATH-derived crystal set (Sillitoe et al., 2021)).

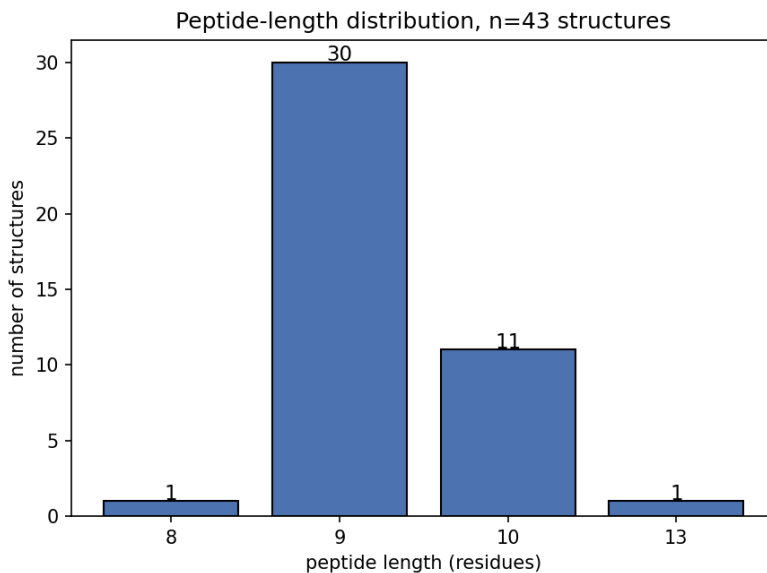


Figure S1: **Peptide length distribution** across all 43 structures in the panel.

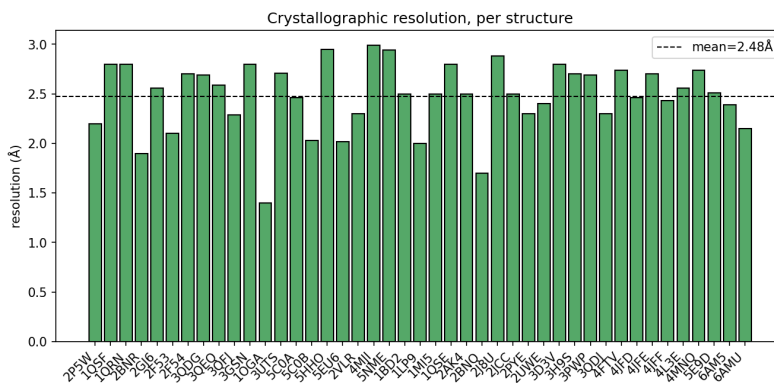


Figure S2: **Crystallographic resolution per structure.**

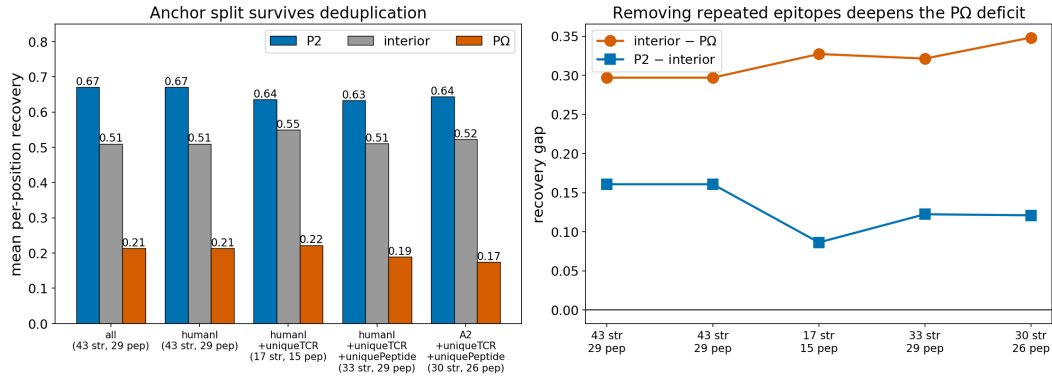


Figure S3: The anchor split is not an artefact of the panel's redundancy. The panel holds 43 structures but only 29 distinct peptides and 22 distinct CDR3 β sequences, so a pooled statistic weights an epitope by how often it was crystallised. Left, P2, interior and PΩ recovery on progressively stricter sets: human MHC class I only, then one structure per TCR family (CDR3 β clustered so clonotypes differing by one or two substitutions collapse), then one per peptide, then restricted to HLA-A*02:01; each group is represented by its best-resolution member, chosen independently of any score. Right, the same result as the two gaps that define it. Both survive, and the PΩ deficit deepens from 0.23 to 0.35 as repeated epitopes are removed. The one exception is P2 on the TCR-only set, which at 17 structures is underpowered rather than contradictory.

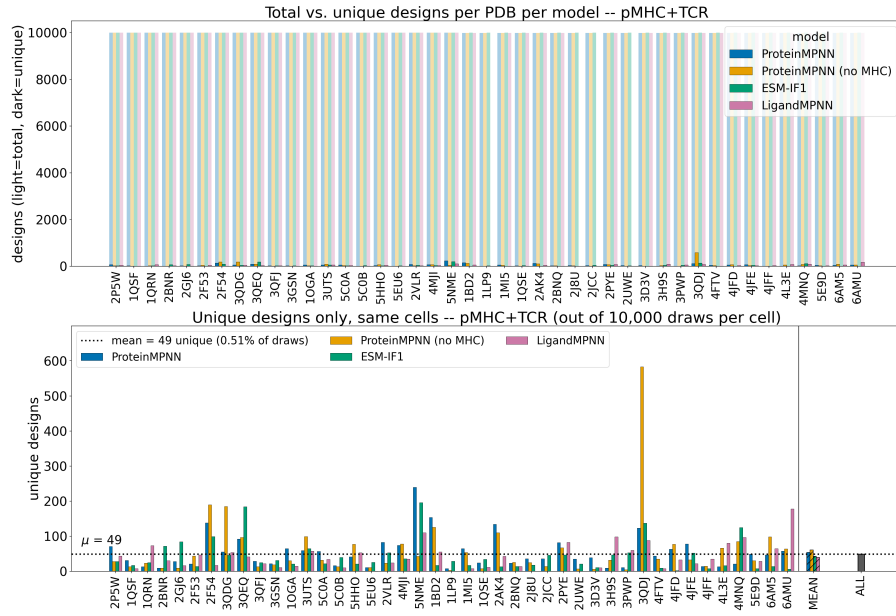


Figure S4: Extreme redundancy at $T=0.1$. 10,000 raw draws per (structure, model); the unique fraction is 0.50% on average and never exceeds a few percent.

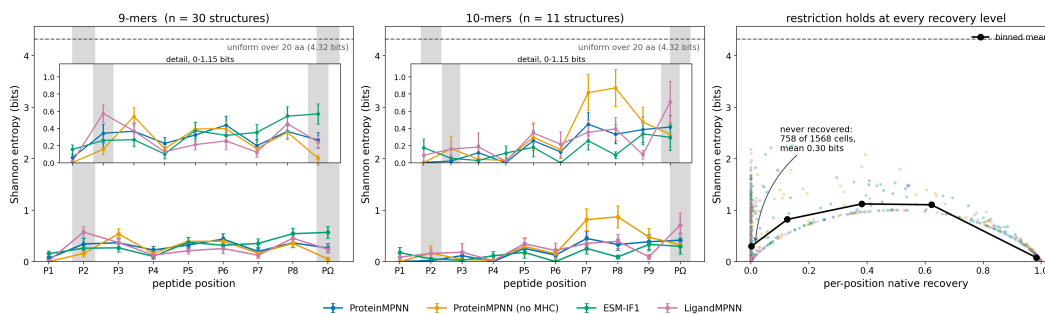


Figure S5: **Per-position sequence diversity of the design distributions**, pooled over the forty-three human MHC class I structures in the pMHC+TCR context. Shannon entropy of the amino-acid distribution each model samples at each peptide position, against the $\log_2 20 = 4.32$ bits a uniform draw over twenty residues would give; anchor positions are shaded, insets rescale to 0–1.15 bits, and error bars are the standard error across structures. Right, the same entropies against per-position native recovery, with the binned mean overlaid: restriction is not a by-product of recovering the native residue, since the positions where the native residue is essentially never sampled are as tightly restricted as the rest.

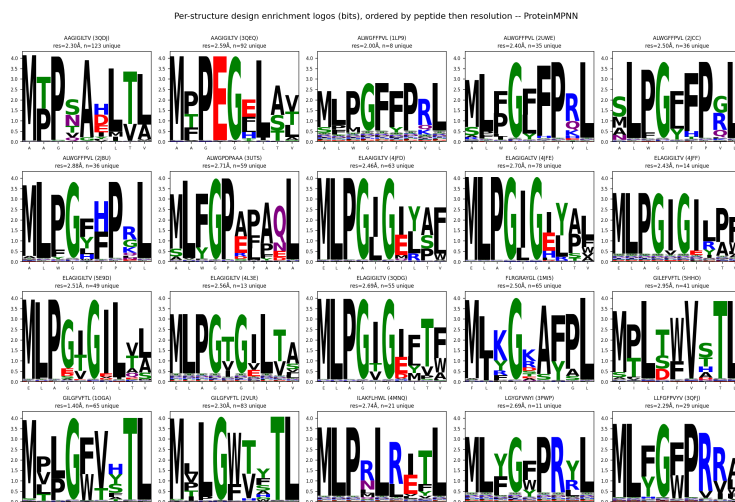


Figure S6: **Per-structure design logos, ProteinMPNN**. All 50 structures, ordered by native peptide then resolution.

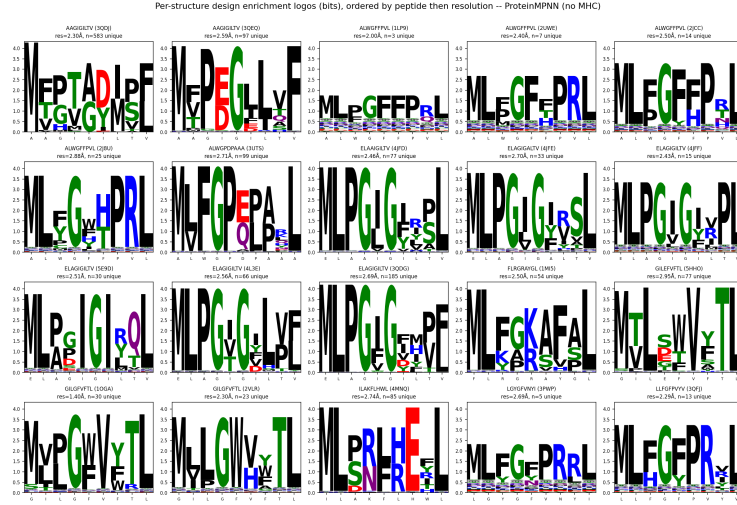


Figure S7: Per-structure design logos, ProteinMPNN (no MHC).

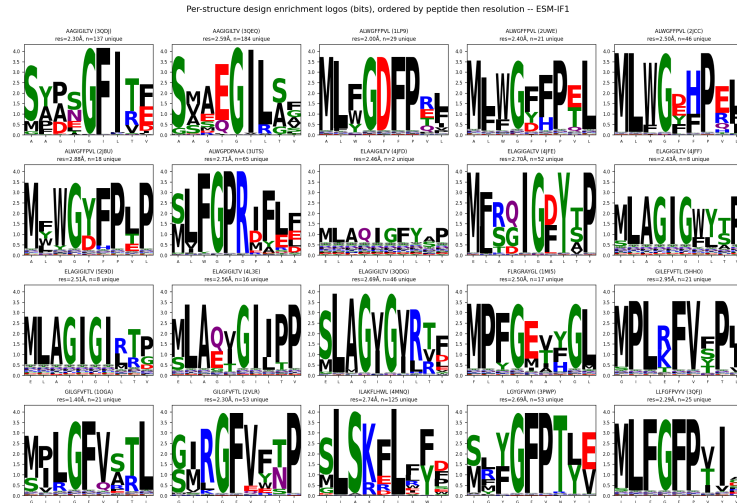


Figure S8: Per-structure design logos, ESM-IF1.

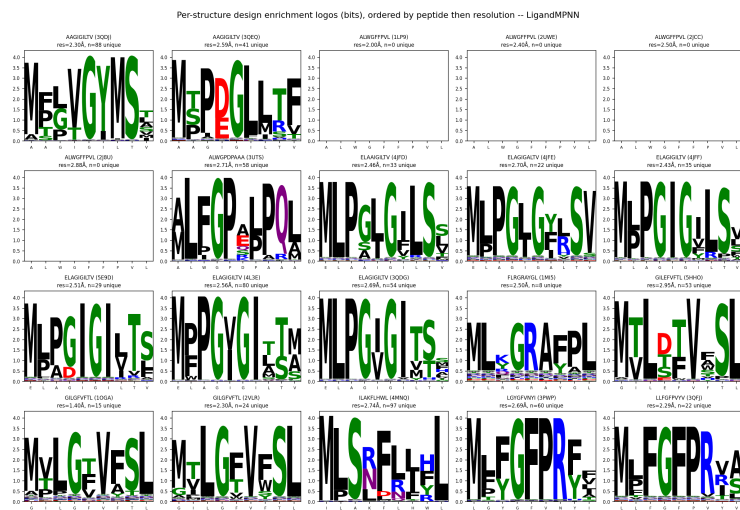


Figure S9: Per-structure design logos, LigandMPNN.

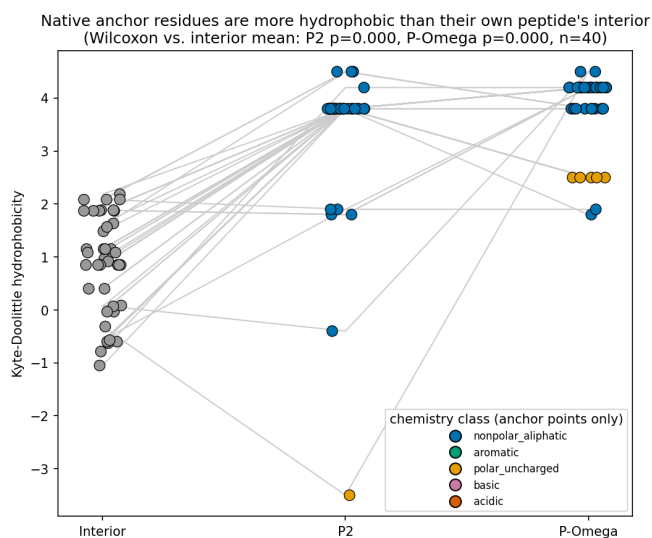


Figure S10: **Both MHC anchors are more hydrophobic than the peptide interior they sit alongside.** Kyte-Doolittle hydrophobicity of the native residue at P2 and P Ω for each of the 40 HLA-A*02:01 structures, plotted against that same peptide's mean over its own interior positions (P3 through the second-to-last, excluding P1). Grey lines connect the three values belonging to one peptide; anchor points are colored by side-chain class. Both anchors sit well above their own interior (paired Wilcoxon, $p < 0.0001$ for both, $n = 40$), so the two pockets do not differ in whether they prefer hydrophobic residues. They differ in how strictly the designs honor that preference, which is the distinction reported in Section 3.2. The points at P Ω colored as polar are chiefly the cysteine-terminated SLLMWITQC/NY-ESO-1 replicates, a residue whose discrete classification understates its hydrophobicity.

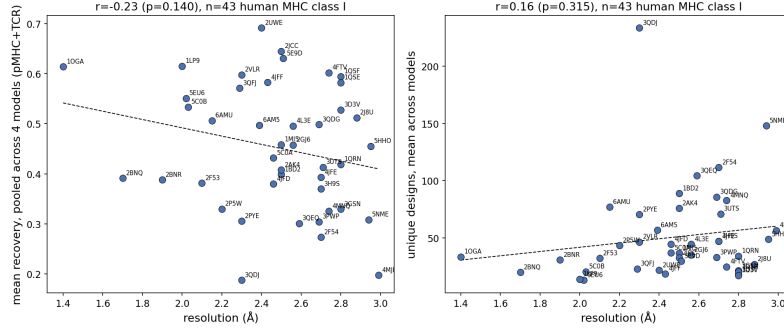


Figure S11: **Resolution and unique-design count vs. mean recovery, pooled across the panel.** $r = -0.23$ ($p = 0.140$) and $r = 0.16$ ($p = 0.315$), $n = 43$ human MHC class I.

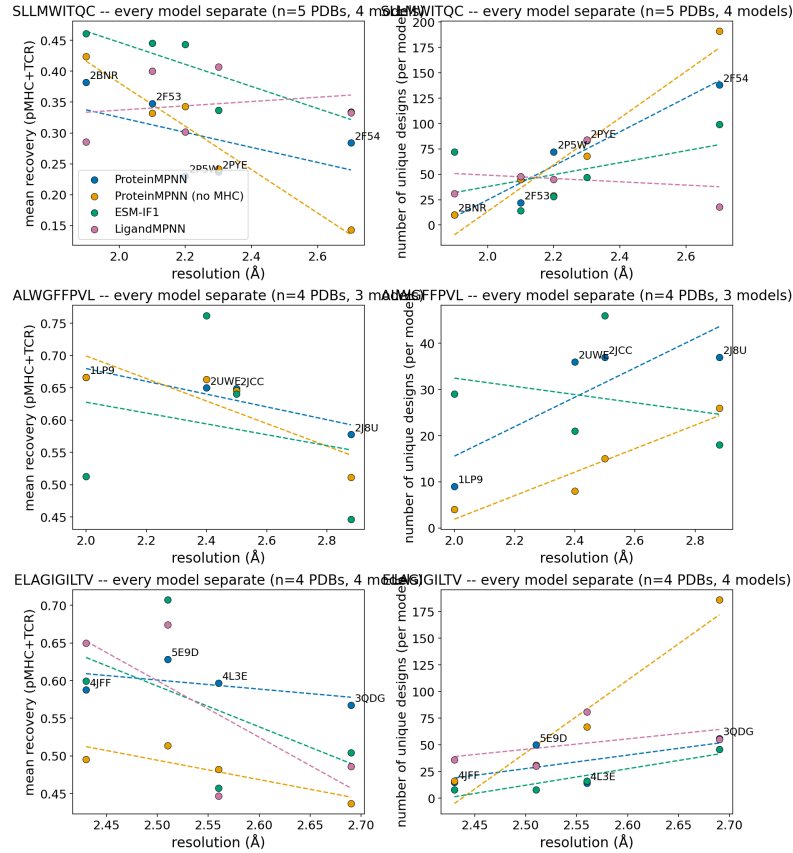


Figure S12: **Resolution dependence in a controlled, same-peptide replicate set.** Four structures (2P5W, 2BNR, 2F53, 2F54), identical native peptide, near-identical TCR clone; every model plotted separately.

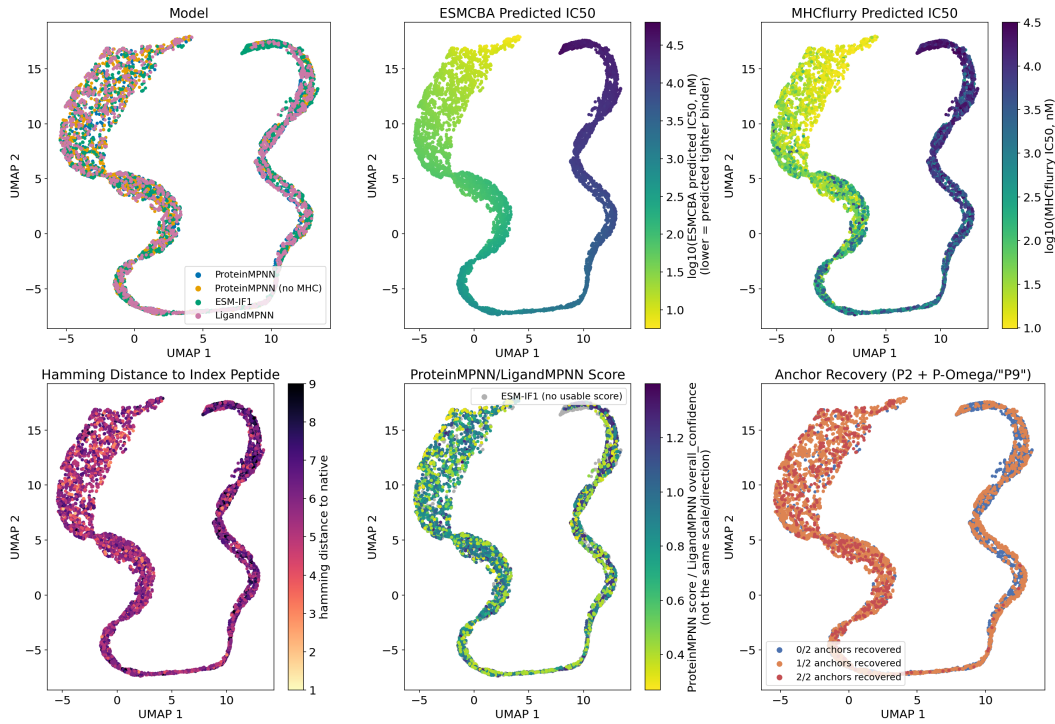


Figure S13: **The sampled sequence space, colored by predicted binding affinity.** One UMAP (ESMCBA embeddings, fit jointly over every unique design), recolored six ways. All four models occupy the same overlapping landscape rather than four separate clusters, and MHCflurry and ESM-CBA, built differently and trained differently, trace the identical gradient across it.

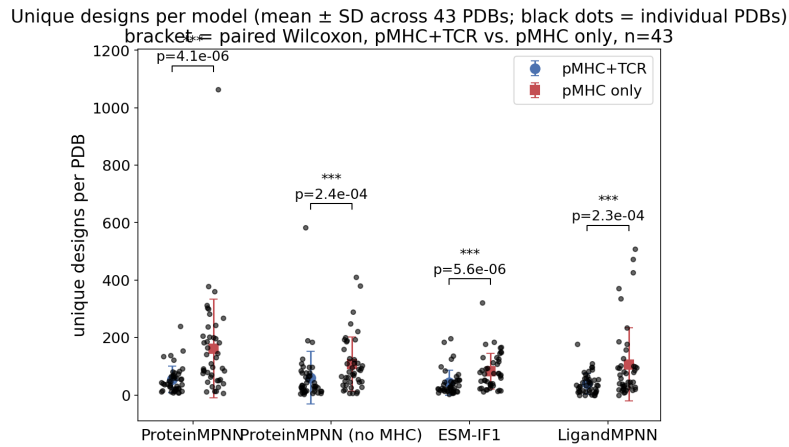


Figure S14: **Unique designs per model, mean $\pm$ SD across structures, pMHC+TCR vs. pMHC only.**

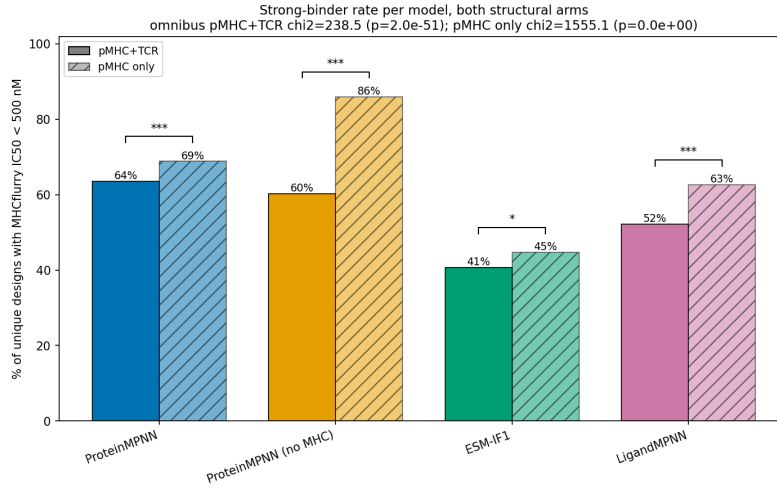


Figure S15: **Removing the TCR raises predicted affinity for every model, by very different amounts.** Strong-binder rate (MHCflurry IC50 < 500 nM) per model in both structural arms; solid bars are pMHC+TCR, hatched bars pMHC only. Brackets give a Fisher's exact test of that model's own shift between arms, Bonferroni-corrected across the four models. Every model shifts up significantly, but the magnitude separates them: ProteinMPNN (no MHC) gains 25.7 percentage points and LigandMPNN 10.5, against 5.3 for ProteinMPNN and 4.0 for ESM-IF1. ESM-IF1 is the lowest-affinity model in both arms. The model ranking is preserved between arms, so the omnibus difference between models (reported above each arm) is not an artifact of which context is used.

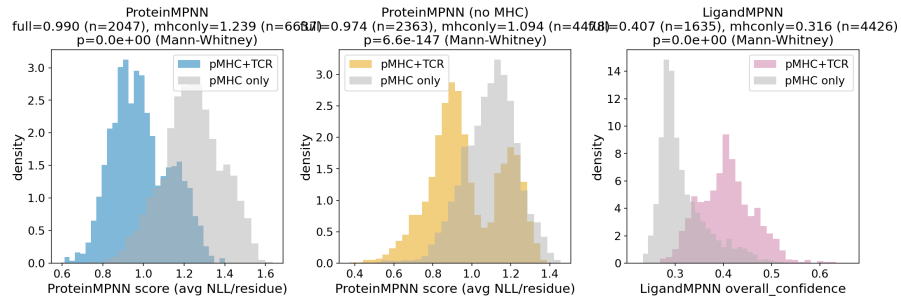


Figure S16: **Each model's own confidence score for its designs shifts when the TCR is removed,** pooled across all forty HLA-A*02:01 structures. ProteinMPNN and ProteinMPNN (no MHC): average per-residue negative log-likelihood, lower is more confident. LigandMPNN: overall confidence, higher is more confident. All three shift toward lower self-assessed confidence without the TCR (Mann-Whitney $p < 10^{-200}$ in all three cases).

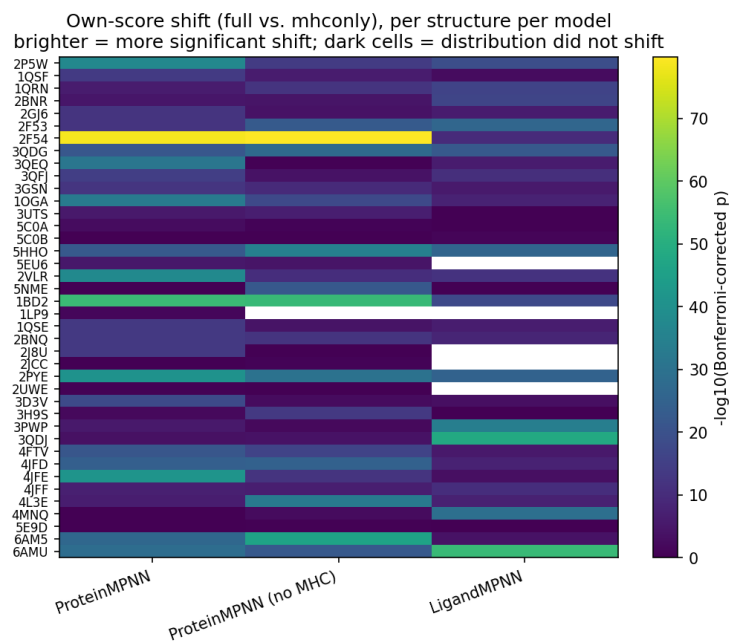


Figure S17: **The own-score shift is not uniform across the panel.** Bonferroni-corrected significance (full vs. mhconly) per structure per model; dark cells are structures where the model's own-score distribution did not shift.

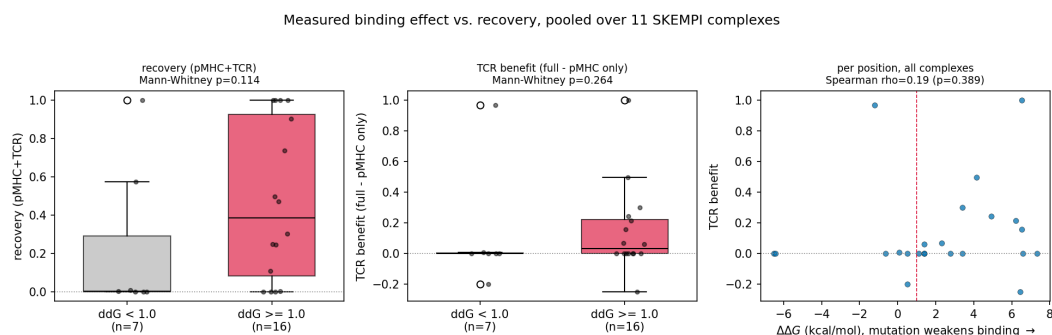


Figure S18: **Measured binding effect vs. recovery, pooled over every SKEMPI complex with designs.** Each point is one (complex, peptide position) with a measured single-mutation $\Delta\Delta G$; positive $\Delta\Delta G$ means the mutation weakens binding, marking a position the interface depends on. Left, recovery with the TCR present. Middle, the recovery gain from TCR context. Right, that gain against $\Delta\Delta G$ directly. 23 positions across 11 complexes, against the five positions in one crystal available before the panel was extended. $\Delta\Delta G$ marks positions the interface depends on, which includes MHC anchors as well as TCR contacts.

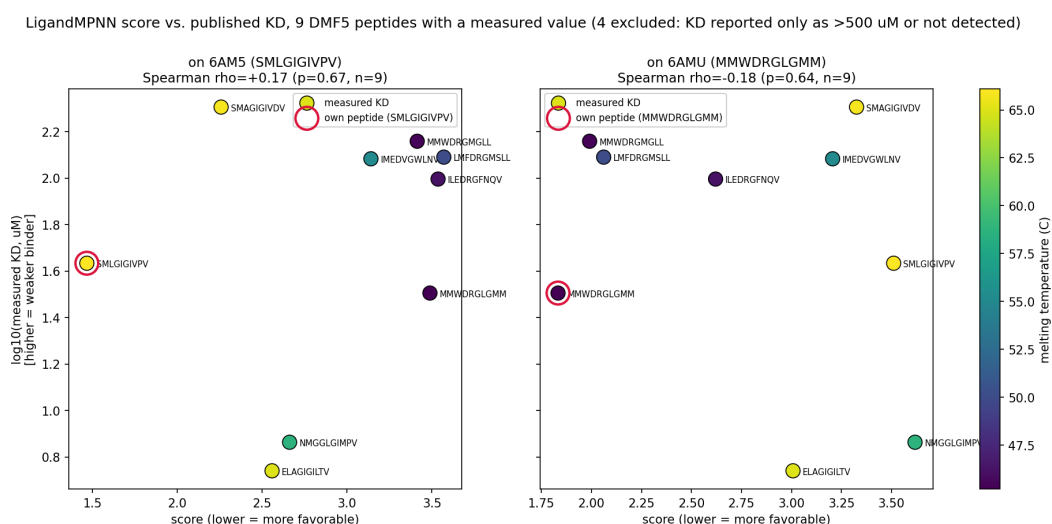


Figure S19: **The score carries no information about measured affinity.** LigandMPNN score against published K_D for the nine DMF5 peptides with an actual measured value, scored on 6AM5 (left) and 6AMU (right); points are colored by melting temperature and each structure's own crystallized peptide is circled. Four further peptides are excluded because their K_D is reported only as > 500 μM or as not detected, which is an assay ceiling rather than a measurement. Neither backbone shows a relationship. Contrast this with Figure 4, where the same scores separate the two peptide families cleanly: these models place a peptide, they do not rank it.